

# Gabriel D. Patrón

**Citizenship:** Canadian, Colombian

**Languages:** English, Spanish

Department of Computing, Imperial College London

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## Professional Appointments

**University of Toronto, Department of Chemical Engineering & Applied Chemistry**

Assistant Professor of Chemical Engineering

Toronto, CA

Starting August 2026

**Imperial College London, Department of Computing, Sargent Centre for Process Systems Engineering**

Research Associate

Advisors: Calvin Tsay (Computing) and Nilay Shah (Chemical Engineering)

London, UK

2024–2026

**University of Waterloo, Department of Chemical Engineering**

Postdoctoral Fellow

Advisor: Luis Ricardez-Sandoval

Waterloo, CA

2023–2024

**National University of Singapore, Department of Chemical and Biomolecular Engineering**

Research Assistant

Advisor: Ning YAN

Singapore, SG

2016

## Education

**University of Waterloo, Department of Chemical Engineering**

PhD, Chemical Engineering

Supervisor: Luis Ricardez-Sandoval

Thesis: "New approaches for the real-time optimization of process systems under uncertainty"

Waterloo, CA

2019–2023

**Imperial College London, Department of Chemical Engineering**

MSc, Chemical Engineering

Supervisor: Amparo Galindo

Thesis: "An Application of Residual Entropy Scaling to Calculate and Predict Viscosity Using the SAFT- $\gamma$  Mie Equation of State"

London, UK

2017–2018

**University of Waterloo, Department of Chemical Engineering**

BASc, Chemical Engineering

Thesis: "Formic Acid-Mediated Pyrolysis of Woody Biomass"

Toronto, CA

2013–2017

## Peer-Reviewed Journal Publications

\*=co-first authors

[J10] Patrón, G.D., Tsay, C., Ricardez-Sandoval, L., 2025. Deep-learning-aided modifier adaptation: synergies with process intensification. *Chemical Engineering and Processing – Process Intensification* 219, 110581.

[J09] Ghilardi, L.M.P., Patrón, G.D., Alcántara, A., Tsay, C., 2025. Integrated design and scheduling of hydrogen processes under uncertainty: a quantile neural network approach. *Industrial & Engineering Chemistry Research*.

[J08] Stordy, B.P.\*, Sepahi, Z.\*, Patrón, G.D., Yang, W., Goodson, A.D., Blackadar, C., Tavares, A.J., Lin, G., Malekjahani, A., Ling, B., Ravichandran, R., Hicks, D.R., Shapiro, M.G. Zhang, M., King, N.P., Baker, D., Ricardez-Sandoval, L.A., Chan, W.C.W., 2025. The Binding Affinities of Serum Proteins to Nanoparticles. *Journal of the American Chemical Society* 147(24), 20475–20492.

[J07] Patrón, G.D., Ricardez-Sandoval, L., 2024. Economically optimal operation of recirculating aquaculture systems under uncertainty. *Computers and Electronics in Agriculture* 220, 108856.

[J06] Patrón, G.D., Toffolo, K., Ricardez-Sandoval, L., 2024. Economic model predictive control for packed bed chemical looping combustion. *Chemical Engineering and Processing – Process Intensification* 198, 109731.

- [J05] Patrón, G.D., Ricardez-Sandoval, L., 2023. Robust real-time optimization and parameter estimation of post-combustion CO<sub>2</sub> capture under economic uncertainty. *Chemical Engineering Science* 281, 119124.
- [J04] Patrón, G.D., Ricardez-Sandoval, L., 2023. Directional modifier adaptation based on input selection for real-time optimization. *Computers & Chemical Engineering* 177, 108351.
- [J03] Patrón, G.D., Ricardez-Sandoval, L., 2022. Low-Variance Parameter Estimation Approach for Real-Time Optimization of Noisy Process Systems. *Industrial & Engineering Chemistry Research* 61(45), 16780–16798.
- [J02] Patrón, G.D., Ricardez-Sandoval, L., 2022. An integrated real-time optimization, control, and estimation scheme for post-combustion CO<sub>2</sub> capture. *Applied Energy* 308, 118302.
- [J01] Patrón, G.D., Ricardez-Sandoval, L., 2020. A robust nonlinear model predictive controller for a post-combustion CO<sub>2</sub> capture absorber unit. *Fuel* 265, 116932.

## Preprints

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- [PP02] Langdon, B., Patrón, G.D., Kappatou, C.D., Lee, R.M., Shafei, B., Qing, J., Misener, R., van der Wilk, M., Tsay, C. Meta-Learning for Sample-Efficient Bayesian Optimisation of Fed-Batch Processes. *arXiv:2605.05382*.
- [PP01] Patrón, G.D., Zhang, D., Ghilardi, L.M.P., Blom, E., Goodridge, M., Solis, E., Hamidreza, J., Angarita, J., Ganesan, N., West, K., Shah, N., Tsay, C. Risk-constrained stochastic scheduling of multi-market energy storage systems. *arXiv:2510.27528*.

## Peer-Reviewed Conference Proceedings

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- [C05, forthcoming] Patrón, G.D., Ganesan, N., Shah, N., Tsay, C. Learnable Decision Trees for Interpretable Energy Arbitrage Scheduling. *IFAC-PapersOnLine*
- [C04, forthcoming] Wang, H., Patrón, G.D., Bo, C., Ricardez-Sandoval, L. Modeling and real-time optimization of an industrial residue oil hydrotreating unit. *IFAC-PapersOnLine*.
- [C03] Patrón, G.D., Ricardez-Sandoval, L., 2024. Bootstrapped gross error detection for efficient and fault-tolerant real-time optimization. *American Control Conference*, 1720–1725.
- [C02] Patrón, G.D., Ricardez-Sandoval, L., 2023. Economic Model Predictive Control of a Recirculating Aquaculture System. *IFAC-PapersOnLine* 56(2); 6156–6161.
- [C01] Patrón, G.D., Ricardez-Sandoval, L., 2020. Real-Time Optimization and Nonlinear Model Predictive Control for a Post-Combustion Carbon Capture Absorber. *IFAC-PapersOnLine* 53(2), 11595–11600.

## Book Chapters

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- [B02, forthcoming] Ronson, D., Ali, O., Patrón, G.D., Ricardez-Sandoval, L., 2026. Advances in Chemical Looping Combustion: Process Integration, Design, and Operations Management. *Process Intensification for Carbon Capture and Utilization*, Elsevier.
- [B01] Patrón, G.D., Ricardez-Sandoval, L., 2026. Online control and optimization for conventional and emerging carbon capture systems. *Encyclopedia of Systems of Systems and Control Engineering* 4, 204–222, Elsevier.

## Oral Presentations (Without Proceedings)

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- [P08] Patrón, G.D., Shah, N., Tsay, C. (2025). Risk-Constrained Two-Stage Demand Response Scheduling for Green Hydrogen Production. *AIChE annual meeting 2025*, 710e.
- [P07] Patrón, G.D., Tsay, C., Ricardez-Sandoval, L. (2025). Deep-Learning-Aided Modifier Adaptation for Real-Time Optimization. *AIChE annual meeting 2025*, 644e.
- [P06] Patrón, G.D., Shah, N., Tsay, C. (2025). Risk-constrained two-stage demand response scheduling for green hydrogen production. *22nd European Conference on Advances in Continuous Optimization*.
- [P05] Patrón, G.D., Shah, N., Tsay, C. (2025). Stochastic Optimization for Demand Response Scheduling of Green Hydrogen Production. *18th INFORMS Computing Society (ICS) Conference*.

[P04] Patrón, G.D., Ricardez-Sandoval, L. (2023). Robust real-time optimization for the long-term economical and sustainable operation of post-combustion carbon capture under uncertainty. 11<sup>th</sup> International Freiberg Conference, Poster 25.

[P03] Patrón, G.D., Ricardez-Sandoval, L. (2022). Partial Modifier Adaptation for Economic Optimization of Process Systems Under Frequent Disturbances and Structural Model Uncertainty. AIChE annual meeting 2022, 434d.

[P02] Patrón, G.D., Ricardez-Sandoval, L. (2022). Parameter Estimation for Real-Time Optimization Under Model Uncertainty and Measurement Noise. AIChE annual meeting 2022, 434g.

[P01] Patrón, G.D., Ricardez-Sandoval, L. (2020). Towards an integrated approach for real-time economic optimization, state estimation, and control for a post-combustion carbon capture absorber section. AIChE annual meeting 2020, 596c.

## Awards and Grants

|                                                                                                             |           |
|-------------------------------------------------------------------------------------------------------------|-----------|
| FIPSE-7 Emerging Star                                                                                       | 2026      |
| Doctoral Thesis Completion Award (\$CAD 5,000)   <b>University of Waterloo</b>                              | 2023      |
| Faculty of Engineering Domestic Doctoral Student Award (\$CAD 28,500 total)   <b>University of Waterloo</b> | 2018–2022 |
| Waterloo Graduate Research Scholarship (\$CAD 68,500 total)   <b>University of Waterloo</b>                 | 2018–2022 |
| Dean's List   <b>University of Toronto</b>                                                                  | 2013–2017 |
| Centre for International Experience Award (\$CAD 1,000)   <b>University of Toronto</b>                      | 2016      |
| Cross-Disciplinary Program Summer Grant (\$CAD 2,000)   <b>University of Toronto</b>                        | 2016      |
| University of Toronto Entrance Scholarship (\$CAD 2,000)   <b>University of Toronto</b>                     | 2013      |

## Teaching and Mentoring

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|-----------------------------------------------------------------------------------------------------------|--------------|
| <b>Graduate teaching</b>                                                                                  |              |
| Imperial Business School                                                                                  | London, UK   |
| TA for MSc in Business Analytics: Deep Learning and Generative AI with Dr. Calvin Tsay                    | 2025         |
| <b>Graduate student supervision</b>                                                                       |              |
| Imperial College London, Department of Computing                                                          | London, UK   |
| PhD candidate, Becky Langdon: “System-aware neural ODE processes for the optimisation of batch processes” | 2024–        |
| University of Waterloo, Department of Chemical Engineering                                                | Waterloo, CA |
| MASc candidate, Dana Ronson: “Modelling of chemical looping combustion with red mud oxygen carrier”       | 2023–        |
| <b>Undergraduate teaching</b>                                                                             |              |
| University of Waterloo, Department of Chemical Engineering                                                | Waterloo, CA |
| TA for CHE420: Introduction to Process Control with Prof. Hector Budman                                   | 2019, 2020   |
| <b>Undergraduate student supervision</b>                                                                  |              |
| University of Waterloo, Department of Chemical Engineering                                                | Waterloo, CA |
| Zhen Ye: “Modifier adaptation for real-time optimization of the Williams-Otto CSTR”                       | 2022         |
| Final year project: “Design of a Chemical Looping Combustion Model for Reducing Carbon Footprint”         | 2021         |
| Final year project: “Modelling and Optimization of Chemical Looping Combustion (CLC) Process”             | 2020         |

## Professional Membership

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|------------------------------------------------------------------------------------------|-------|
| <b>American Institute of Chemical Engineers (AIChE):</b> Post-doctoral Researcher Member | 2022– |
| <b>Canadian Society for Chemical Engineering (CSCeE):</b> Postdoctoral Fellow Member     | 2022– |
| <b>International Federation of Automatic Control (IFAC):</b> Affiliate Member            | 2022– |

## Academic Service

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| <b>Journal reviewer:</b> AIChE Journal, Applied Intelligence, The Canadian Journal of Chemical Engineering, Computers and Electronic in Agriculture, Digital Chemical Engineering, Engineering Applications of Artificial Intelligence, Industrial & Engineering Chemistry Research, Journal of Process Control. |  |
| <b>Conference reviewer:</b> American Controls Conference (ACC), Dynamics and Control of Process Systems (DYCOPS).                                                                                                                                                                                                |  |
| <b>Conference session organizer:</b> 22nd Conference on Advances in Continuous Optimization.                                                                                                                                                                                                                     |  |

## Industrial Experience

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### **EllisDon Corporation**

2015

M.E.I.T. Intern, New Oakville Trafalgar Memorial Hospital

- Worked with specialty teams - Mechanical, Electrical, and Information Technology - during the commissioning of the project.
- Performed calibration, testing, and troubleshooting for hospital communications and emergency systems - to meet strict hospital regulations and standards.
- Identified system deficiencies and liaised with subcontractors to find solutions.
- Modified drawings for hoarding permit applications using Autodesk.

### **EllisDon Corporation**

2014

Estimating Intern

- Was a part of the proposal team that formed an estimate and won the Eglinton Light Rail Transit (ELRT) project.
- Management of several project-specific tender packages, including assessment of requirements based on specifications, qualification process, management of quotes, quantity takeoffs, and estimates.